

12.2 Viewing PCA from an Ellipse Perspective



Core skills you will master in this section:

- ▶ Use ellipses or ellipsoids to intuitively describe the geometric shape of a data distribution.
- ▶ Center each column of data (subtract the mean) so that the centroid moves to the origin.
- ▶ The covariance matrix describes the structure of the data.
- ▶ Spectral decomposition extracts the data's principal-axis directions and variances.
- ▶ The trace of a square matrix is the sum of all elements on its main diagonal.
- ▶ Project the data onto the eigenvector directions to achieve information compression.
- ▶ Rotate the coordinate axes (the principal-component directions) to "straighten" the shape of the data.

Imagine the data matrix as a cloud, and a suitable geometric shape for describing this cloud can be an ellipse or an ellipsoid. Principal component analysis is about finding a suitable coordinate system in which to "straighten" the ellipse or ellipsoid that represents the data, and then completing the projection in the new coordinate system.

⚠ Note that when using principal component analysis, we assume the data follows a multivariate Gaussian distribution. Probability and Statistics Is Not Hard, another book in the "Math Is Not Hard" series, covers the multivariate Gaussian distribution in detail.

Can we find a new set of coordinate axes that "straightens" the data and "flattens" it along the most useful direction? This is the geometric intuition behind principal component analysis.

We use the data matrix to compute the covariance matrix; we use the covariance matrix to describe the ellipse or ellipsoid. We use eigenvalue decomposition (spectral decomposition) to find the principal axes (the principal-component directions) of the ellipse or ellipsoid. We use orthogonal projection to reduce the data's dimensionality onto the principal-component directions. Following this line of thinking, we will be using tools introduced earlier in this book: spectral decomposition, orthogonal projection, and the Gram matrix.

Steps of Principal Component Analysis

The steps of principal component analysis are as follows:

- ▶ Center the columns of the data matrix (standardize them if the variances differ greatly across features);
- ▶ Compute the covariance matrix;

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Code and PDF file downloads: <https://github.com/Visualize-ML>

The companion video lectures for this book are all published on Bilibili — 生姜 DrGinger: <https://space.bilibili.com/513194466>

Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

- ▶ Perform eigenvalue decomposition on the covariance matrix;
- ▶ Arrange the corresponding eigenvectors in order of decreasing eigenvalue;
- ▶ Select the principal-component direction (the vector corresponding to the largest eigenvalue);
- ▶ Project the data matrix onto the principal-component direction.

⚠ Note that when computing the covariance matrix, data centering is (in most cases) already included.

Let's now go through principal component analysis step by step.

The Data Matrix

Suppose we have a set of two-dimensional data points, with the corresponding data matrix

$$\mathbf{X}_{300 \times 2} = \begin{bmatrix} 2.272 & 1.672 \\ 2.031 & 3.272 \\ \vdots & \vdots \\ 4.894 & 3.716 \end{bmatrix}$$

The data matrix has 300 rows, that is, 300 sample points.

As shown in Figure 1, the scatter of data points looks like a tilted, elliptical cloud.

➡ Figure 1's ellipse is actually a Mahalanobis-distance contour. The Mahalanobis distance is a topic covered in Chapter 13, Section 4 of this book.

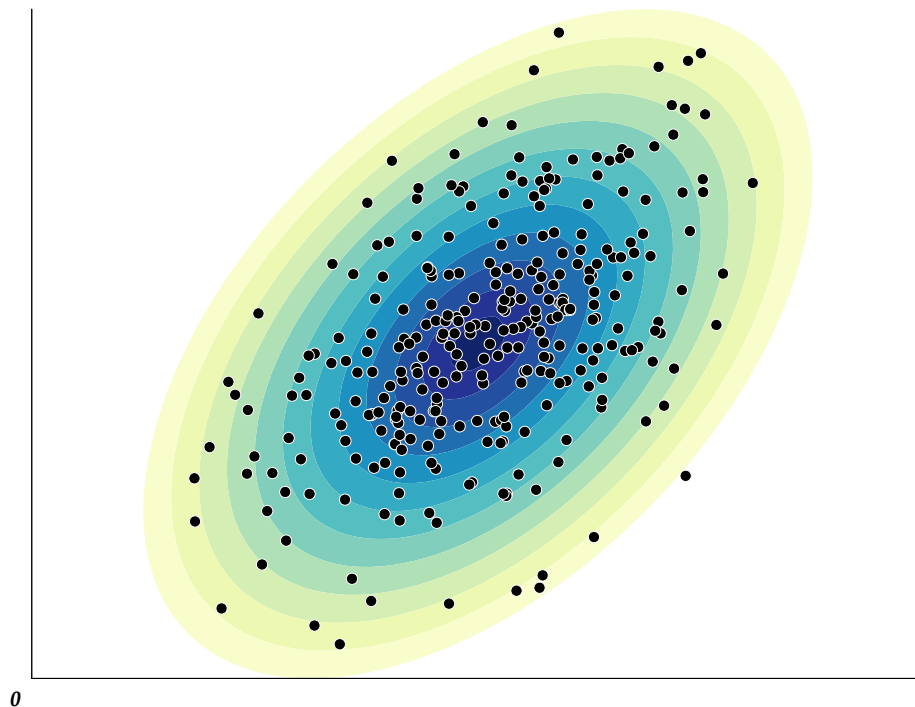


Figure 1. Rotated Elliptical "Data Cloud"

As shown in Figure 2, the centroid of the data is the center of the rotated ellipse.

Through the following matrix multiplication, we compute the specific location of the centroid

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \frac{\mathbf{X}^T \mathbf{1}}{n} = \frac{1}{300} \times \begin{bmatrix} 2.272 & 1.672 \\ 2.031 & 3.272 \\ \vdots & \vdots \\ 4.894 & 3.716 \end{bmatrix}^T @ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

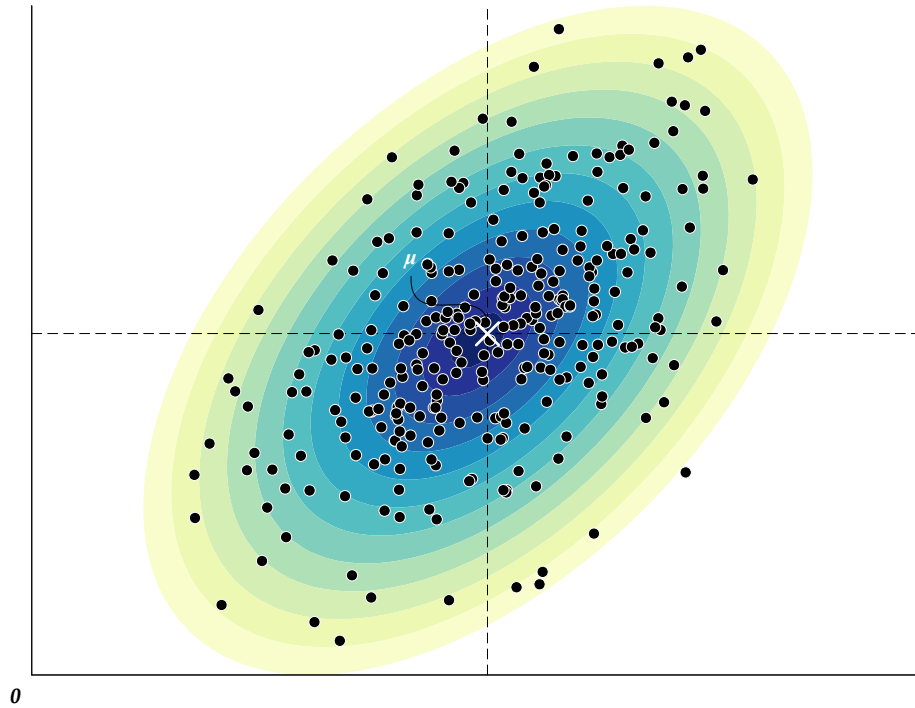


Figure 2. The Centroid of the "Data Cloud"

As shown in Figure 3, the data matrix $\mathbf{X}$ projected onto the horizontal axis gives $\mathbf{x}_1$, that is

$$\mathbf{X} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = [\mathbf{x}_1 \quad \mathbf{x}_2] @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{x}_1$$

The centroid $\boldsymbol{\mu}$ is projected onto the x_1 axis, giving μ_1

$$\mu_1 = \boldsymbol{\mu}^T \mathbf{e}_1 = \mathbf{e}_1^T \boldsymbol{\mu} = [1 \quad 0] @ \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

The equation above is essentially a scalar projection.

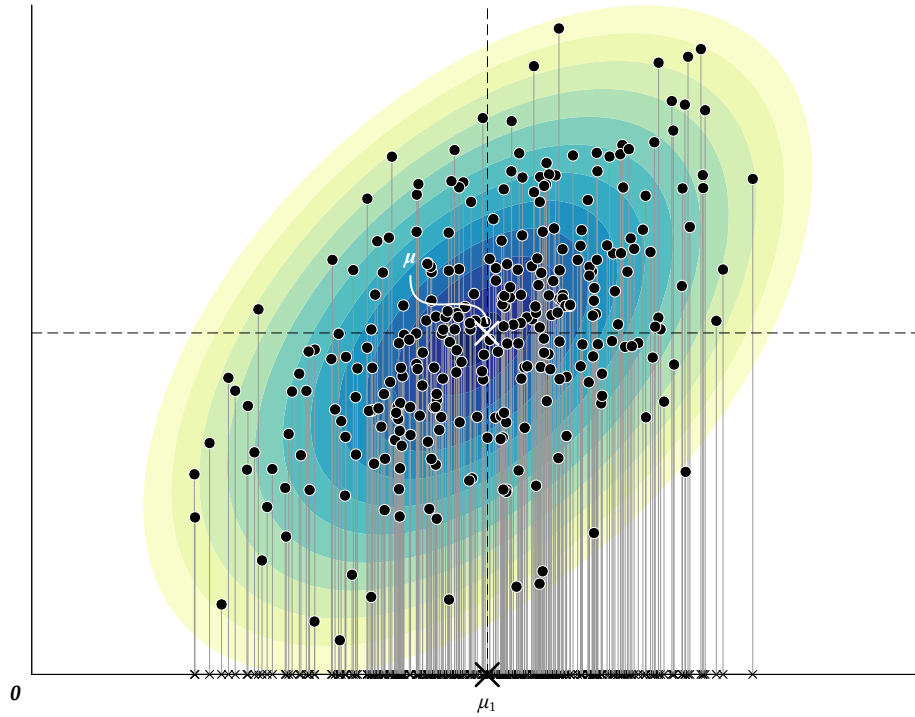


Figure 3. Data Matrix Projected onto the Horizontal Axis

Similarly, as shown in Figure 4, the data matrix $\mathbf{X}$ projected onto the vertical axis gives $\mathbf{x}_2$, that is

$$\mathbf{X} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [\mathbf{x}_1 \quad \mathbf{x}_2] @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \mathbf{x}_2$$

The centroid $\boldsymbol{\mu}$ is projected onto the x_2 axis, giving μ_2

$$\mu_2 = \boldsymbol{\mu}^T \mathbf{e}_2 = \mathbf{e}_2^T \boldsymbol{\mu} = [0 \quad 1] @ \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

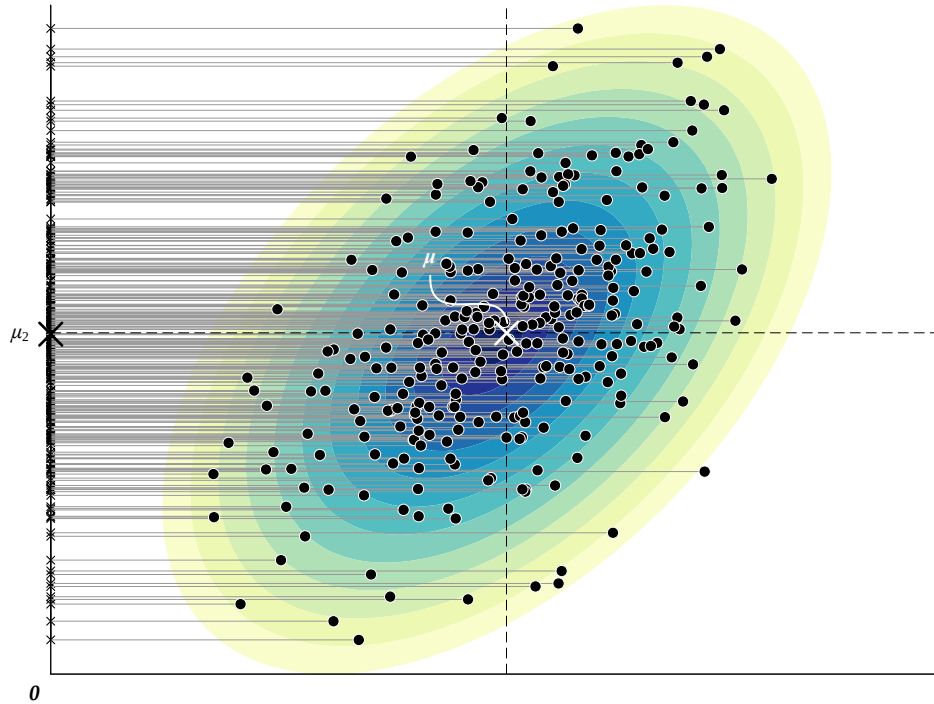


Figure 4. Data Matrix Projected onto the Vertical Axis

Data Centering

As mentioned in the previous section, data centering (removing the mean) means that each column of the data matrix X has its column mean subtracted, giving X_c . Using the broadcasting rule

$$X_c = X - \mu^T$$

As shown in Figure 5, (column) centering does not change the shape of the data's distribution, but it does shift the data's center to the origin (the zero vector $\mathbf{0}$).

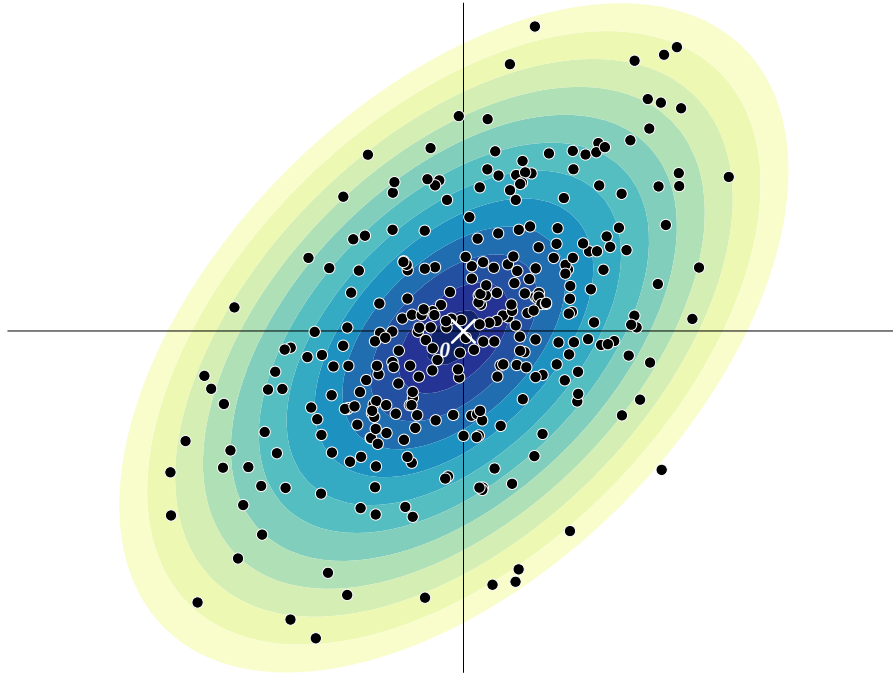


Figure 5. Data Centering

The Covariance Matrix

Using the centered data, we compute the covariance matrix among the features, which represents the linear correlations among the variables.

As mentioned in the previous section, the covariance matrix is essentially a type of Gram matrix of the centered data matrix, that is

$$\Sigma = \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1}$$

Substituting in the specific values, we get Figure 1's data covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

The main diagonal of the covariance matrix consists of variances; that is, the variance of the data projected onto the different axes in Figure 3 and Figure 4 is all 1.

Taking the square root of the variance gives the standard deviation, which is also 1 in each case.

The off-diagonal elements of the covariance matrix are covariances, which means the covariance between $\mathbf{x}_1$ and $\mathbf{x}_2$ is 0.5.

$\mathbf{x}_1$ and $\mathbf{x}_2$'s linear correlation coefficient is also 0.5, that is

$$\rho_{1,2} = \frac{\text{cov}(\mathbf{x}_1, \mathbf{x}_2)}{\sigma_1 \sigma_2} = \frac{0.5}{1 \times 1} = 0.5$$

⚠ Note that, because Figure 1's data variances (standard deviations) do not differ greatly, we can perform principal component analysis simply by applying eigenvalue decomposition directly to the

covariance matrix; however, if the variances differ greatly, the data must first be standardized — a topic covered in the next section.

Eigenvalue Decomposition

Let's perform eigenvalue decomposition on the covariance matrix computed above

$$\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix} 1.5 & 0 \\ 0 & 0.5 \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

In the equation above, what do $\mathbf{V}$ and $\mathbf{\Lambda}$ actually mean?

Let's analyze this using a geometric perspective, as always.

As shown in Figure 6, the centered data is projected in the direction of $\mathbf{v}$, and the resulting projected data $\mathbf{y}$ is

$$\mathbf{y} = \mathbf{X}_c \mathbf{v}$$

Since $\mathbf{X}_c$'s centroid is located at the origin, $\mathbf{y}$'s mean is 0.

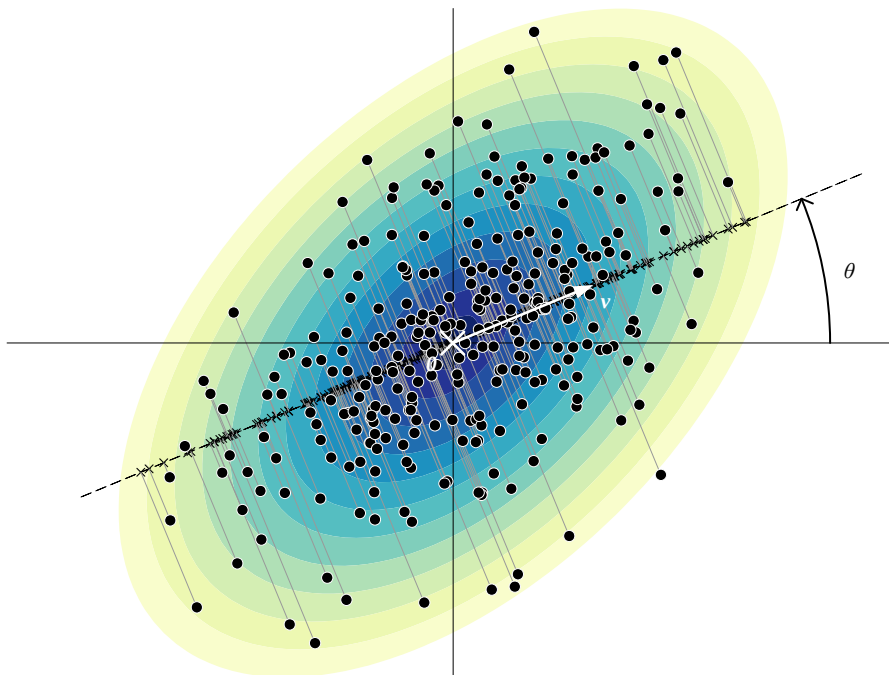


Figure 6. Projection of the Centered Data

Compute the projected data $\mathbf{y}$'s variance

$$\text{var}(\mathbf{y}) = \frac{\mathbf{y}^T \mathbf{y}}{n-1} = \frac{(\mathbf{X}_c \mathbf{v})^T \mathbf{X}_c \mathbf{v}}{n-1} = \mathbf{v}^T \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1} \mathbf{v} = \mathbf{v}^T \Sigma \mathbf{v}$$

Vector $\mathbf{v}$'s angle with the positive direction of the horizontal axis is θ .

Figure 7 shows how the variance changes with θ , and the maximum value of the equation above corresponds to the covariance matrix Σ 's largest eigenvalue (1.5); the minimum value corresponds to the smallest eigenvalue (0.5).

➔ The equation above is a special Rayleigh quotient. Also, have you noticed that Figure 7 is a trigonometric function?! These are all topics discussed in Chapter 13, Section 3 of this book.

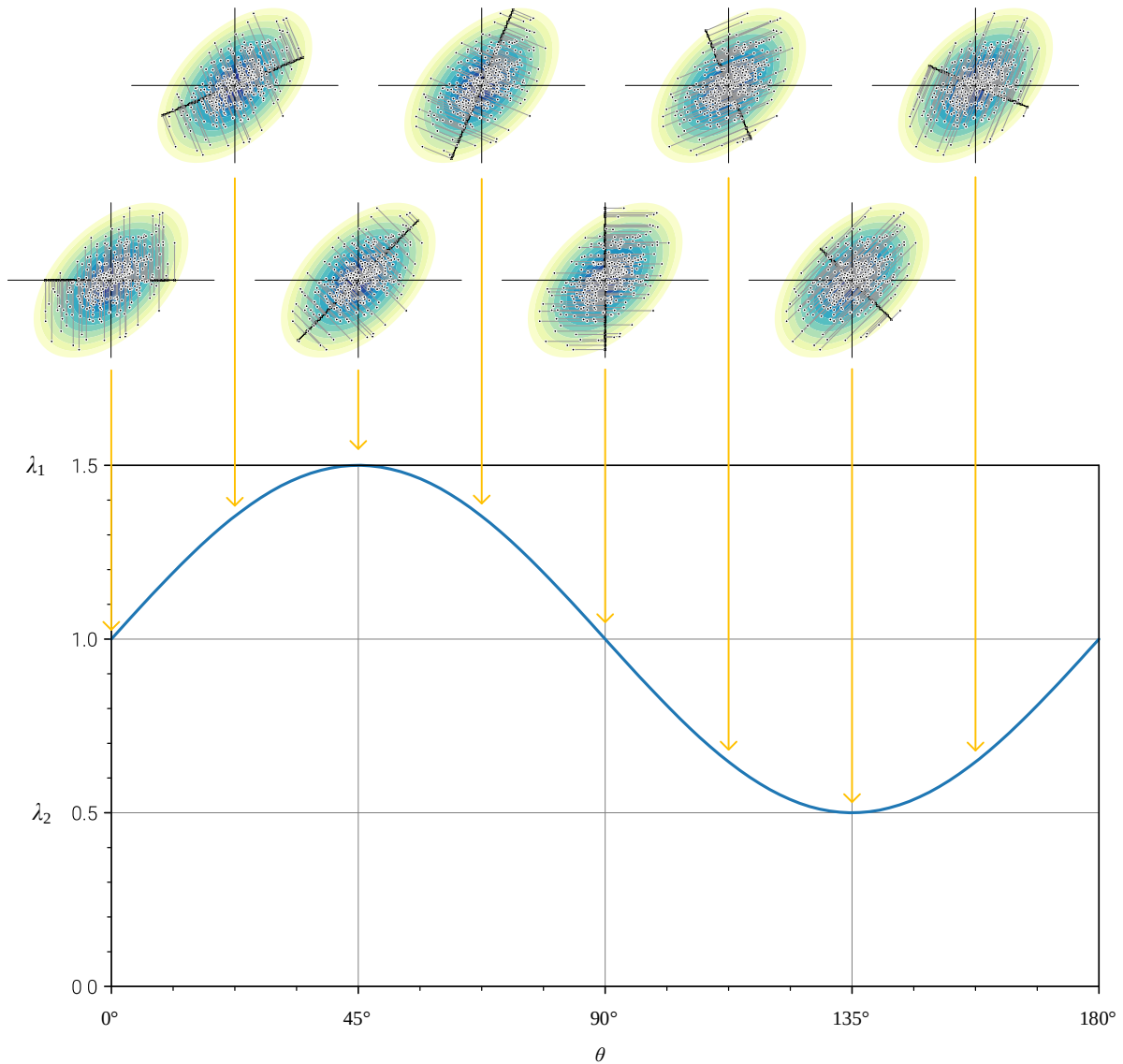


Figure 7. Variance as a Function of θ

And $\mathbf{V}$'s two eigenvectors are exactly the directions corresponding to the largest and smallest eigenvalues,

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2] = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$\mathbf{v}_1$ and $\mathbf{v}_2$'s angles with the positive direction of the horizontal axis are 45 degrees and 135 degrees, respectively – corresponding to the angles at Figure 7's maximum and minimum values.

According to the equation above, the covariance matrix can be diagonalized

$$\mathbf{V}^T \Sigma \mathbf{V} = \mathbf{A}$$

Writing $\mathbf{V}$ as the column vectors $\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix}$ and expanding the equation above, we get

$$\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix}^T \Sigma \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \end{bmatrix} \Sigma \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1^T \Sigma \mathbf{v}_1 & \mathbf{v}_1^T \Sigma \mathbf{v}_2 \\ \mathbf{v}_2^T \Sigma \mathbf{v}_1 & \mathbf{v}_2^T \Sigma \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

Please pay special attention to the following two equations – we'll need them shortly.

$$\mathbf{v}_1^T \Sigma \mathbf{v}_1 = \lambda_1$$

$$\mathbf{v}_2^T \Sigma \mathbf{v}_2 = \lambda_2$$

Also worth noting: the trace of the covariance matrix equals the trace of the $\mathbf{A}$ obtained from eigenvalue decomposition, that is

$$\begin{aligned} \text{trace}(\Sigma) &= \text{trace}\left(\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}\right) = 1 + 1 = 2 \\ \text{trace}(\mathbf{A}) &= \text{trace}\left(\begin{bmatrix} 1.5 & 0 \\ 0 & 0.5 \end{bmatrix}\right) = 1.5 + 0.5 = 2 \end{aligned}$$

Simply put, the trace of a square matrix equals the sum of all its eigenvalues (counting multiplicities).

Projection

The larger the eigenvalue, the greater the variance of the data along that direction, and the richer the information.

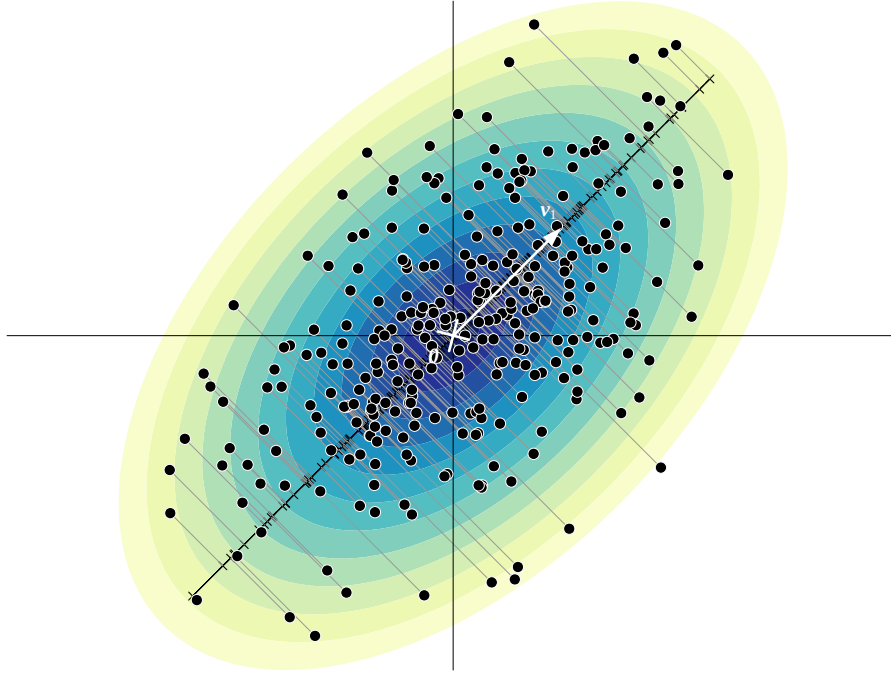
As shown in Figure 8, the centered data matrix $\mathbf{X}_c$ projected onto $\mathbf{v}_1$ gives the projected data $\mathbf{y}_1$

$$\mathbf{y}_1 = \mathbf{X}_c \mathbf{v}_1$$

Compute $\mathbf{y}_1$'s variance

$$\text{var}(\mathbf{y}_1) = \frac{\mathbf{y}_1^T \mathbf{y}_1}{n-1} = \frac{(\mathbf{X}_c \mathbf{v}_1)^T \mathbf{X}_c \mathbf{v}_1}{n-1} = \mathbf{v}_1^T \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1} \mathbf{v}_1 = \mathbf{v}_1^T \Sigma \mathbf{v}_1 = \lambda_1$$

This means $\mathbf{y}_1$'s variance is 1.5 (λ_1).

Figure 8. Data Projected onto $\mathbf{v}_1$

Similarly, as shown in Figure 9, the centered data matrix $\mathbf{X}_c$ projected onto $\mathbf{v}_2$ gives the projected data $\mathbf{y}_2$

$$\mathbf{y}_2 = \mathbf{X}_c \mathbf{v}_2$$

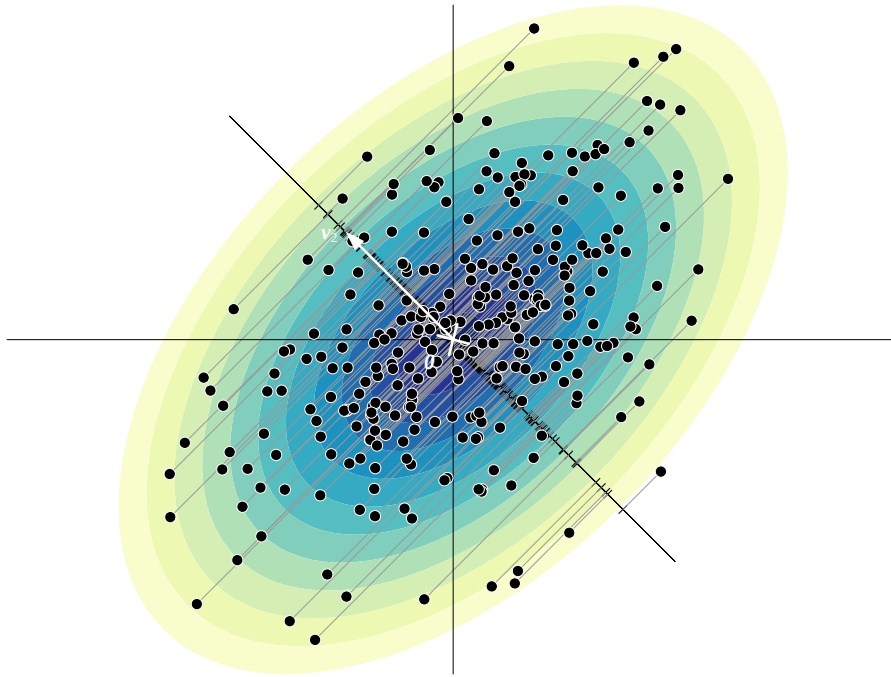
Compute $\mathbf{y}_2$'s variance

$$\text{var}(\mathbf{y}_2) = \frac{\mathbf{y}_2^T \mathbf{y}_2}{n-1} = \frac{(\mathbf{X}_c \mathbf{v}_2)^T \mathbf{X}_c \mathbf{v}_2}{n-1} = \mathbf{v}_2^T \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1} \mathbf{v}_2 = \mathbf{v}_2^T \boldsymbol{\Sigma} \mathbf{v}_2 = \lambda_2$$

This means $\mathbf{y}_2$'s variance is 0.5 (λ_2).

Looking at Figure 8 and Figure 9, we find that $\mathbf{v}_1$ corresponds to the ellipse's major-axis direction, which is also the direction of the first principal component; $\mathbf{v}_2$ corresponds to the ellipse's minor-axis direction, which is also the direction of the second principal component.

Projecting the data in different directions gives different projection results, corresponding to different distributions; projecting onto the ellipse's major-axis direction gives the largest standard deviation of the projected data, while projecting onto the minor-axis direction gives the smallest.

Figure 9. Data Projected onto v_2

Viewing the Data in the Orthonormal Basis V

Since the covariance matrix is a symmetric matrix, the V obtained from eigenvalue decomposition (spectral decomposition) is an orthogonal matrix. V 's column vectors form an orthonormal basis.

Looking at it from another angle, as shown in Figure 10, principal component analysis is nothing more than viewing the same set of data from within V .

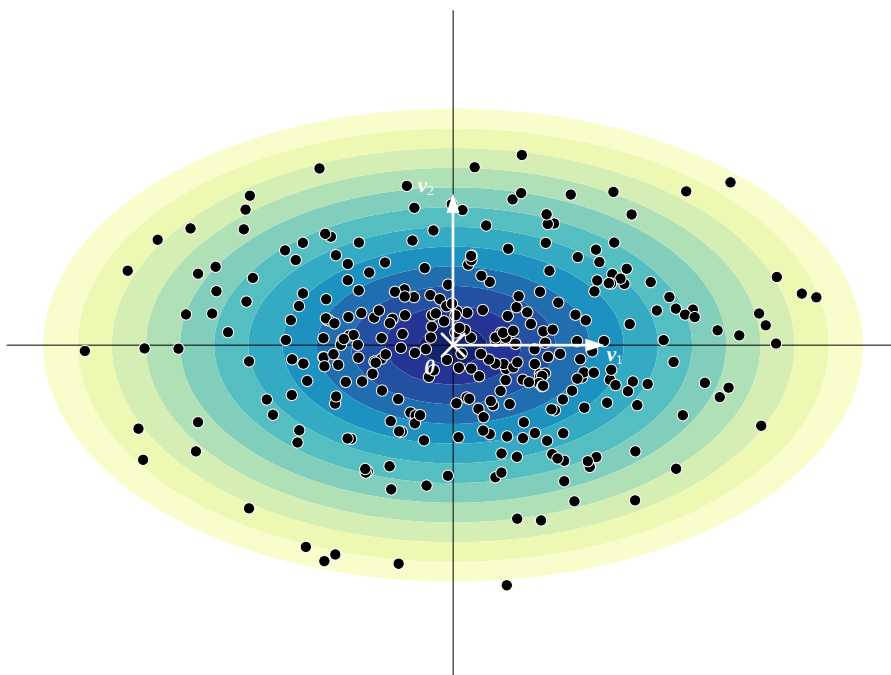
Figure 10. In $[v_1, v_2]$, Viewing the Centered Data

Figure 11 shows, within the basis $[\mathbf{v}_1, \mathbf{v}_2]$, the projection result onto $\mathbf{v}_1$. Figure 12 shows, within the basis $[\mathbf{v}_1, \mathbf{v}_2]$, the projection result onto $\mathbf{v}_2$.

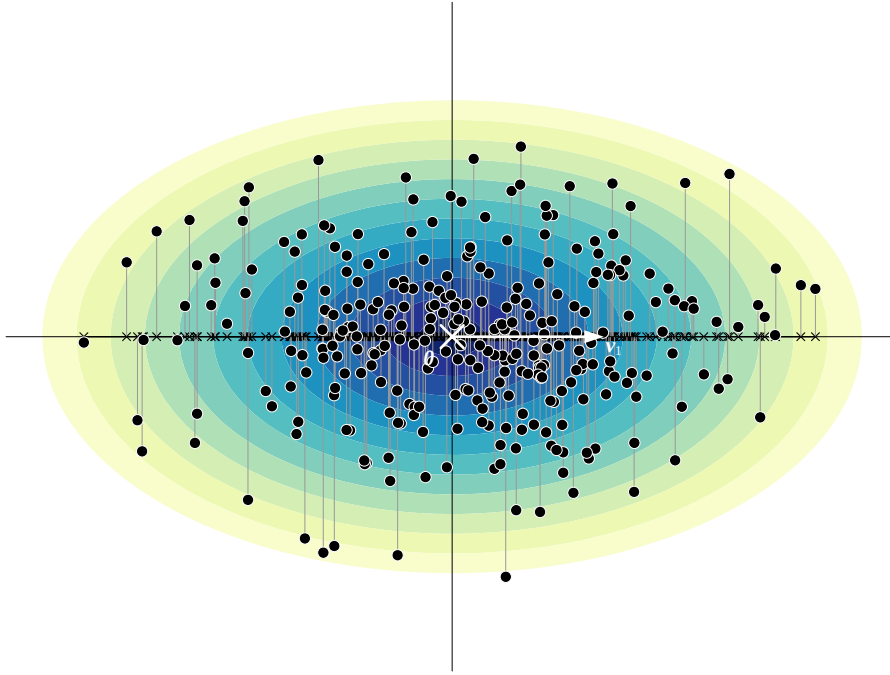


Figure 11. In $[\mathbf{v}_1, \mathbf{v}_2]$, Viewing the Centered Data, Projected onto $\mathbf{v}_1$

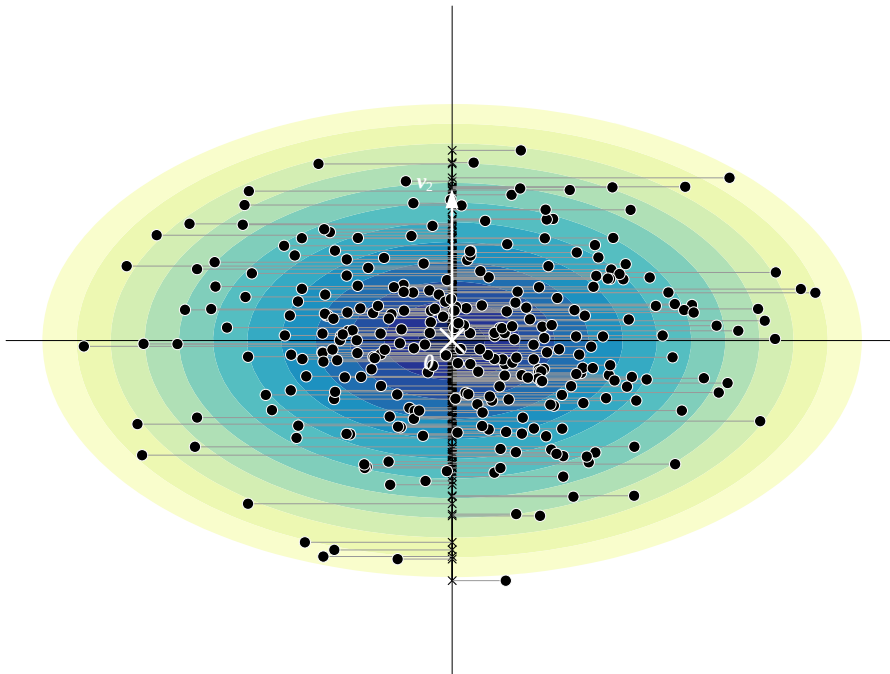


Figure 12. In $[\mathbf{v}_1, \mathbf{v}_2]$, Viewing the Centered Data, Projected onto $\mathbf{v}_2$



Please study LA_12_02_01.ipynb on your own — it completes the main computations and visualizations for this section.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please familiarize yourself with the Mahalanobis distance.

Q2. Please study on your own the `mahalanobis_distance()` function in LA_12_02_01.ipynb, which computes the Mahalanobis distance.

Q3. Please rewrite the visualization function in LA_12_02_01.ipynb as a custom function for repeated use.